TBWB

Angular Advanced
White Label / Multi tenant-solutions

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White Label / Multi tenant-solutions

Single core app; various routes, features, services, UI, etc per client.

Whitelabel / multi tenant solutions

- Desire:
  - Having one (1) "Core" application with basic functionality, layout, navigation etc.
  - Specific clients can have specific Features, Routes, UI, Logic, Service, etc, based on their specific needs.
  - Every client has its own "feature pack".
- Question:

"*What would be the best way to organize such a solution?*"

Typical Separation of this architecture

- Core application:
  - Shell/layout, skeleton app, authentication, http interceptors, design system (preferably in themes), generic features
- Customer Features:
  - Every client has its own package of features and additions
  - Extra pages, extra routes, other feature flags, different config layouts, other branding/theme, etc.
- Important!
  - "Variation" here is extra config, or plug-in modules.
  - NOT `if customer ===‘XYZ’` statements in the code.

Dynamic loading – options per customer

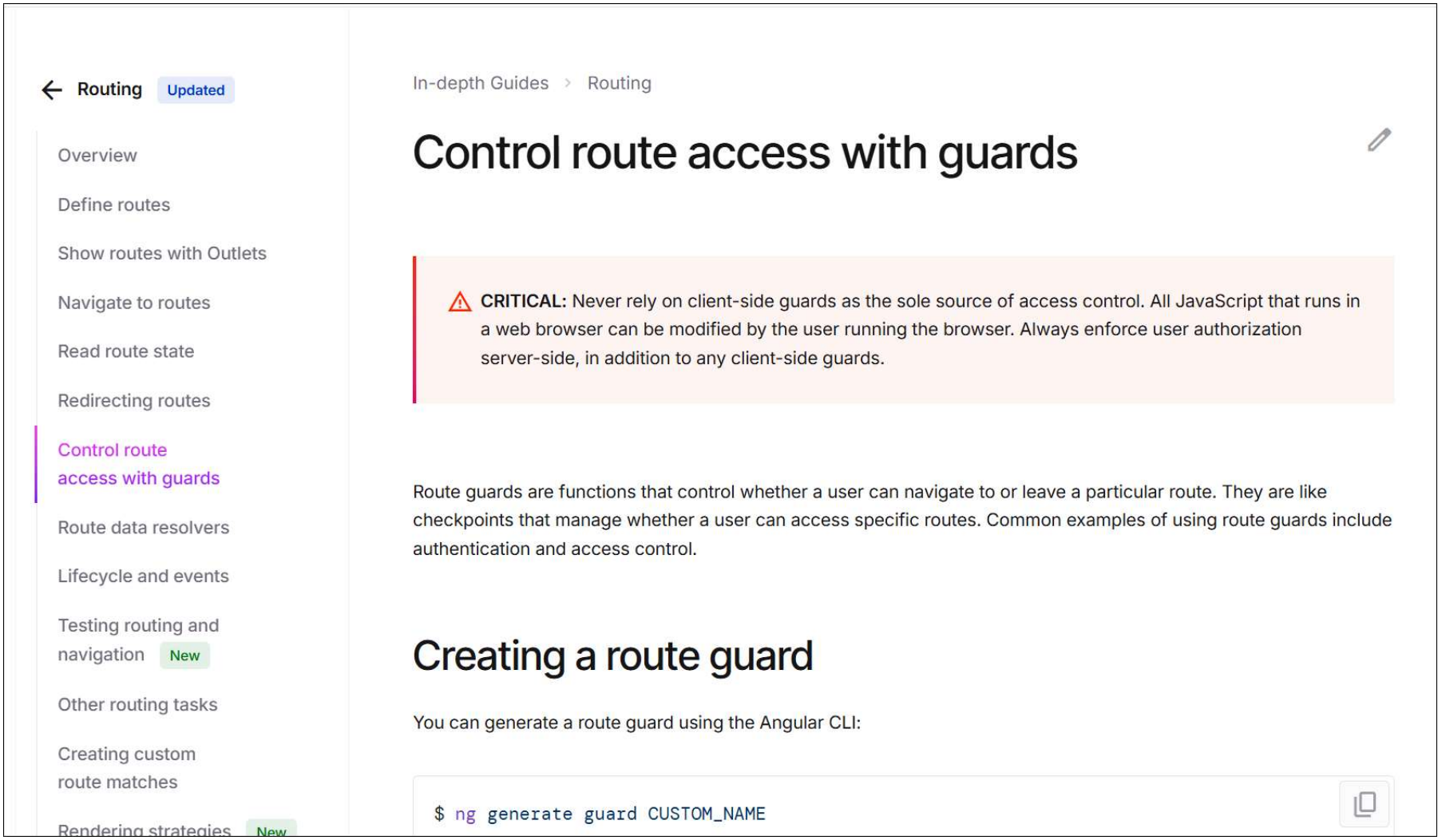
- Goal: "*keep initial bundle small AND do not ship irrelevant code to customers*".

- Basically three (3) levels available:
  1. Build-time selection (simple, but multiple builds)
  2. Runtime selection within a single build (most common)
  3. Truly dynamic plugins/microfrontends (heavy, but super flexible)

Roughly three options for white labeling

1. Build-time selection (simple, but multiple builds)
    1. file replacements / env / customer-specific builds
    2. This works, but you lose the *'build once, deploy everywhere'* advantage
2. Runtime selection within a single build (most common)
    1. runtime config determines which routes/features are 'on'
    2. lazy loading via router (`loadChildren` / `loadComponent`)
    3. possibly *guards* (`canMatch`) to make routes conditionally matchable (useful for feature flags)
3. Truly dynamic plugins/microfrontends (heavy, but super flexible)
    1. Module Federation / Native Federation: customer plugins can even be deployed separately
    2. useful if customers have completely different feature sets or teams deploy separately

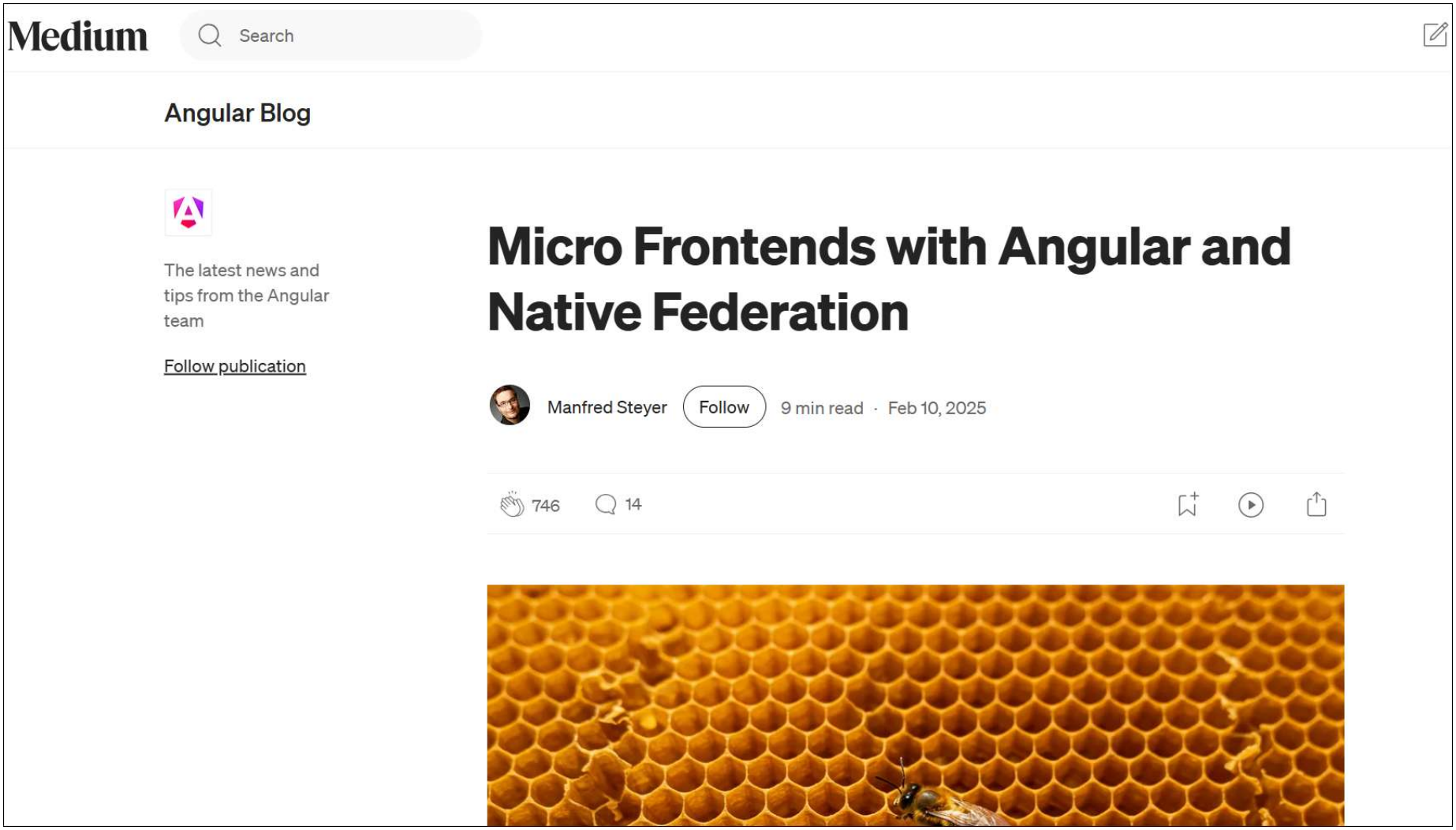
Some background info: route guards



https://angular.dev/guide/routing/route-guards

Micro frontends and Module federation

(warning: heavy stuff!)



https://blog.angular.dev/micro-frontends-with-angular-and-native-federation-7623cfc5f413

Example: ‘feature packs’ per customer

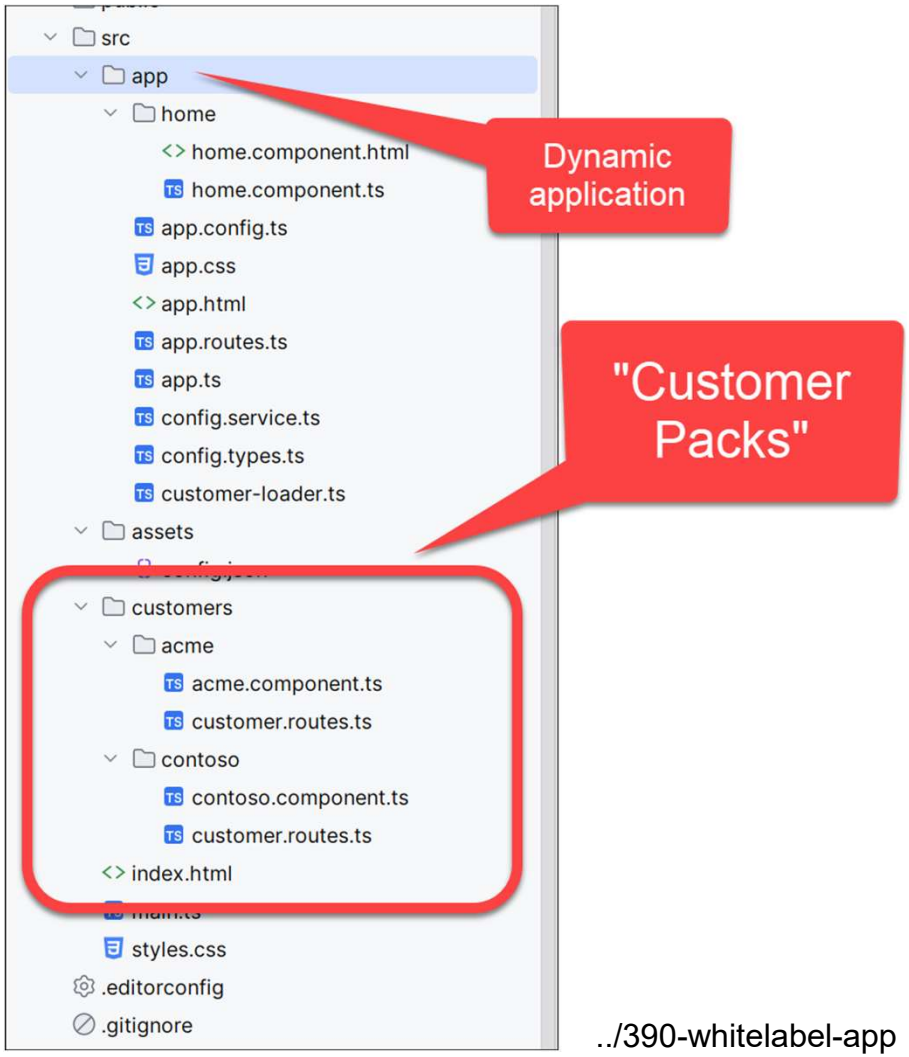
- Example solution:

Every customer has one (1) lazy-loaded route bundle (acting as a “customer feature pack”) which is only loaded once the customer is active.

This means in code...

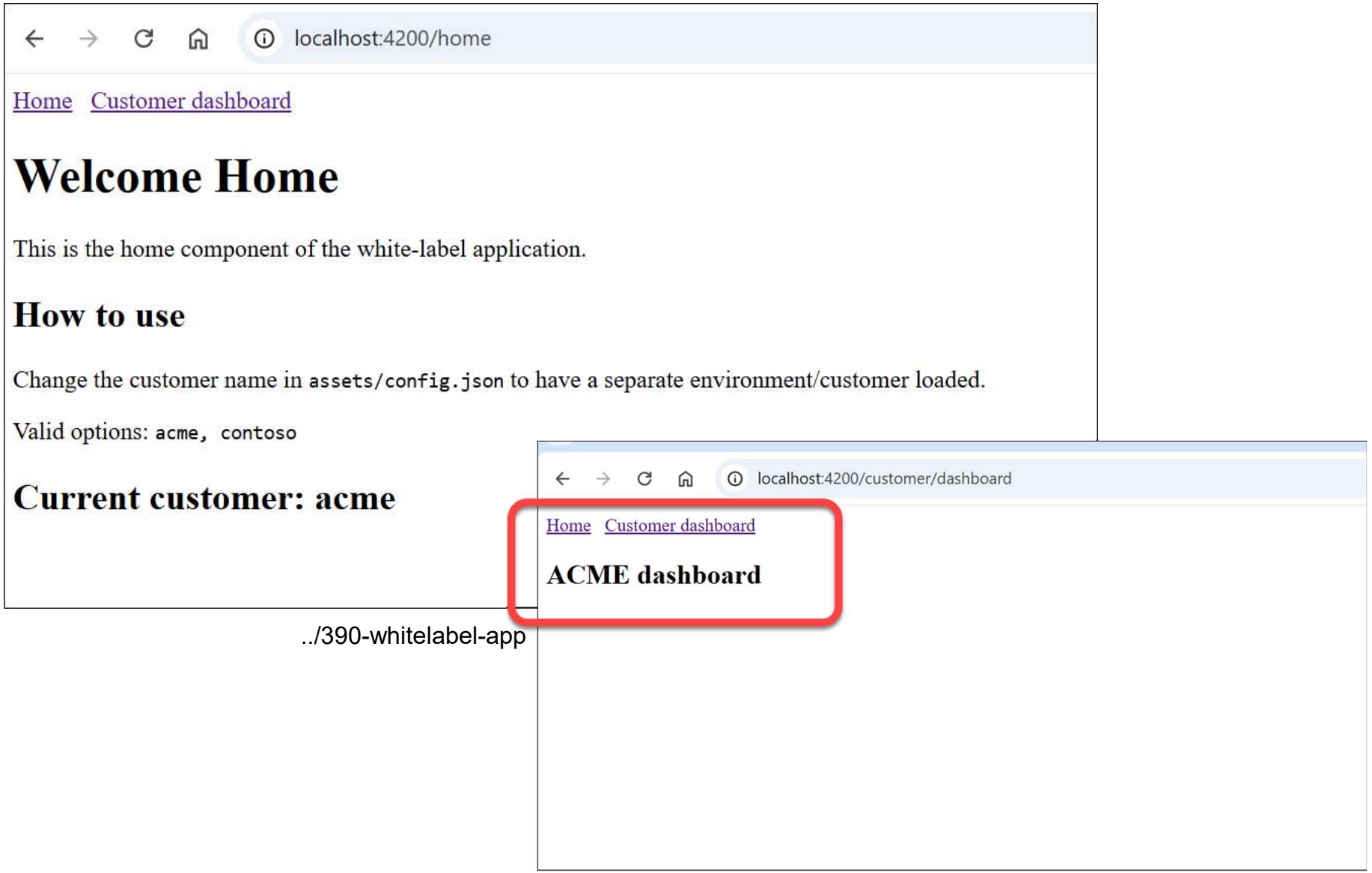
- There is one core app
- There are multiple customer folders
- each customer has their own `routes.ts` export
- selection occurs at runtime (via config, like in dynamic applications)
- Lazy loading (via `import()`) in main `app.routes.ts`.

App Structure / architecture



../390-whitelabel-app

When customer `acme` is active:



../390-whitelabel-app

assets/config.json + config.types.ts

- Decide which customer + which features are being loaded:

```json
{
  "customer": "acme",
  "validCustomers": ["acme", "contoso"],
  "features": {
    "apiUrl": "https://api.test.com",
    "featureX": true,
    "featureY": false
  }
}
```

```typescript
//config.types.ts
export interface AppRuntimeConfig {
  customer: string;
  validCustomers: string[];
  features: {
    apiUrl: string;
    featureX: boolean;
    featureY: boolean;
  };
}
```

`config.service.ts` (see also “dynamic applications”)

```
//config.service.ts
import { Injectable } from '@angular/core’;
…

@Injectable({ providedIn: 'root' })
export class ConfigService {
  private _config!: AppRuntimeConfig;

  get config(): AppRuntimeConfig {
    if (!this._config) throw new Error('Config not loaded yet');
    return this._config;
  }

  constructor(private http: HttpClient) {}

  async load(): Promise<void> {
    const url = `assets/config.json?v=${Date.now()}`;
    this._config = await firstValueFrom(this.http.get<AppRuntimeConfig>(url));
  }
}
```

customer.loader.ts

(providing a map of customer routes, called by `provideRouter`)

```ts
//customer-loader.ts
import {Routes} from '@angular/router';

export type CustomerKey = 'acme' | 'contoso';

const CUSTOMER_ROUTES: Record<CustomerKey, () => Promise<Routes>> = {
  acme: () => import('../customers/acme/customer.routes')
    .then(m => m.CUSTOMER_ROUTES),
  contoso: () => import('../customers/contoso/customer.routes')
    .then(m => m.CUSTOMER_ROUTES),
};

export function loadCustomerRoutes(customer: CustomerKey): Promise<Routes> {
  return CUSTOMER_ROUTES[customer]();
}
```

app.routes.ts (basic, core routes)

```
// app.routes.ts
import { Routes } from '@angular/router';

export const CORE_ROUTES: Routes = [
  { path: '', pathMatch: 'full', redirectTo: 'home' },
  { path: 'home', loadComponent: () =>
import('./home/home.component').then(m => m.HomeComponent) },

  // placeholder: is being replaced during init!
  { path: 'customer', children: [] },

  { path: '**', redirectTo: 'home' },
];
```

app.config.ts

```
// app.config.ts
import { loadCustomerRoutes } from './customer-loader’;
…
export const appConfig: ApplicationConfig = {
  providers: [
    provideHttpClient(),
    provideRouter(CORE_ROUTES), // using the CORE_ROUTES placeholder

    provideAppInitializer(async () => {
       …
      const customerRoutes = await loadCustomerRoutes(customer);

      // The magic happens here
      const next = CORE_ROUTES.map(r =>
        r.path === 'customer' ? { ...r, children: customerRoutes } : r
      );

      // reset router
      router.resetConfig(next);
    }),
  ],
};
```

Routes per customer:

```
// customer.routes.ts :: customer ACME
import { Routes } from '@angular/router';
import { AcmeComponent } from './acme.component';

export const CUSTOMER_ROUTES: Routes = [
  { path: 'dashboard', component: AcmeComponent },
];
```

```
// customer.routes.ts :: customer CONTOSO
import { Routes } from '@angular/router';
import {ContosoComponent} from './contoso.component';

export const CUSTOMER_ROUTES: Routes = [
  { path: 'dashboard', component: ContosoComponent },
];
```

Components + other stuff per customer

```
// AcmeComponent
import { Component } from '@angular/core';

@Component({
  standalone: true,
  template: `<h2>ACME dashboard</h2>`,
})
export class AcmeComponent {}
```

```
// ContosoComponent
import { Component } from '@angular/core';

@Component({
  standalone: true,
  template: `<h2>CONTOSO dashboard</h2>`,
})
export class ContosoComponent {}
```

This example is NOT:

- Microfrontends
- Module federation
- Multiple builds
- runtime eval / dynamic path string
- if/else customer checks in components

Signals when to look further

Signal	Next steps
Multiple teams per customer	Module Federation
Independent deploys	Microfrontends
Third Party Plugins	Remote MFEs
Runtime URLs per tenant	Native Federation

Questions

- Q: Do I distribute / build all customer artifacts?
- *A: Yes, You distribute one build artefact (the `dist/` output). That build does contain code for all customer packs (as separate lazy chunks). But runtime config determines which chunk(s) are ever retrieved.*
- **Often: distribute one /build or Docker image, with specific config.json's**

Per customer separate location?

- Publishing is twice, but building is once!
  - `https://acme.example.com/` → same build, own `config.json`
  - `https://contoso.example.com/` → same build, own `config.json`

More concerns

- Q: “But then I ship customer code to everyone... is that okay?”
- A: “That's the trade-off with this option”
  - Pro: 1 build, simple, little infrastructure
  - Con: you ‘ship’ all customer code (although it's only downloaded when you navigate to it)
- In many B2B SaaS scenarios, that's fine, because it's client-side code anyway (no secrets)
  - the extra bytes are mainly in chunks that are never loaded. You save a lot of build/deploy complexity
- When is it NOT okay?
  - If customers are absolutely not allowed to see each other's UI/features (compliance/contractual)
  - if bundles become too large
  - you really want to deploy different teams/versions per customer to: